# Atlantic Hurricane Activity

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```
library(readr)

storms <- read_csv("storms.csv", show_col_types = FALSE)

head(storms)
```

```
## # A tibble: 6 x 13
##   name   year month   day  hour   lat  long status       category  wind pressure
##   <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <chr>           <dbl> <dbl>    <dbl>
## 1 Amy    1975     6    27     0  27.5 -79   tropical de~       NA    25     1013
## 2 Amy    1975     6    27     6  28.5 -79   tropical de~       NA    25     1013
## 3 Amy    1975     6    27    12  29.5 -79   tropical de~       NA    25     1013
## 4 Amy    1975     6    27    18  30.5 -79   tropical de~       NA    25     1013
## 5 Amy    1975     6    28     0  31.5 -78.8 tropical de~       NA    25     1012
## 6 Amy    1975     6    28     6  32.4 -78.7 tropical de~       NA    25     1012
## # i 2 more variables: tropicalstorm_force_diameter <dbl>,
## #   hurricane_force_diameter <dbl>
```

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following object is masked _by_ '.GlobalEnv':
##
##     storms
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(tidyr)
library(ggplot2)

# Part 1: Missing Value Analysis
```

```r
missing_counts <- storms |>
  summarise(across(everything(), ~ sum(is.na(.x)))) |>
  pivot_longer(
    everything(),
    names_to = "variable",
    values_to = "n_missing"
  ) |>
  filter(n_missing > 0)

print(missing_counts)
```

```
## # A tibble: 3 x 2
##   variable                     n_missing
##   <chr>                            <int>
## 1 category                         14734
## 2 tropicalstorm_force_diameter      9512
## 3 hurricane_force_diameter          9512
```

```r
# Part 2: Custom Function 1
# Count unique hurricanes by year
annual_hurricane_counts <- function(df) {
  df |>
    group_by(year, name) |>
    summarise(
      is_hurricane = any(status == "hurricane"),
      is_cat5 = any(category == 5, na.rm = TRUE),
      .groups = "drop"
    ) |>
    group_by(year) |>
    summarise(
      hurricanes = sum(is_hurricane),
      category5 = sum(is_cat5),
      .groups = "drop"
    )
}

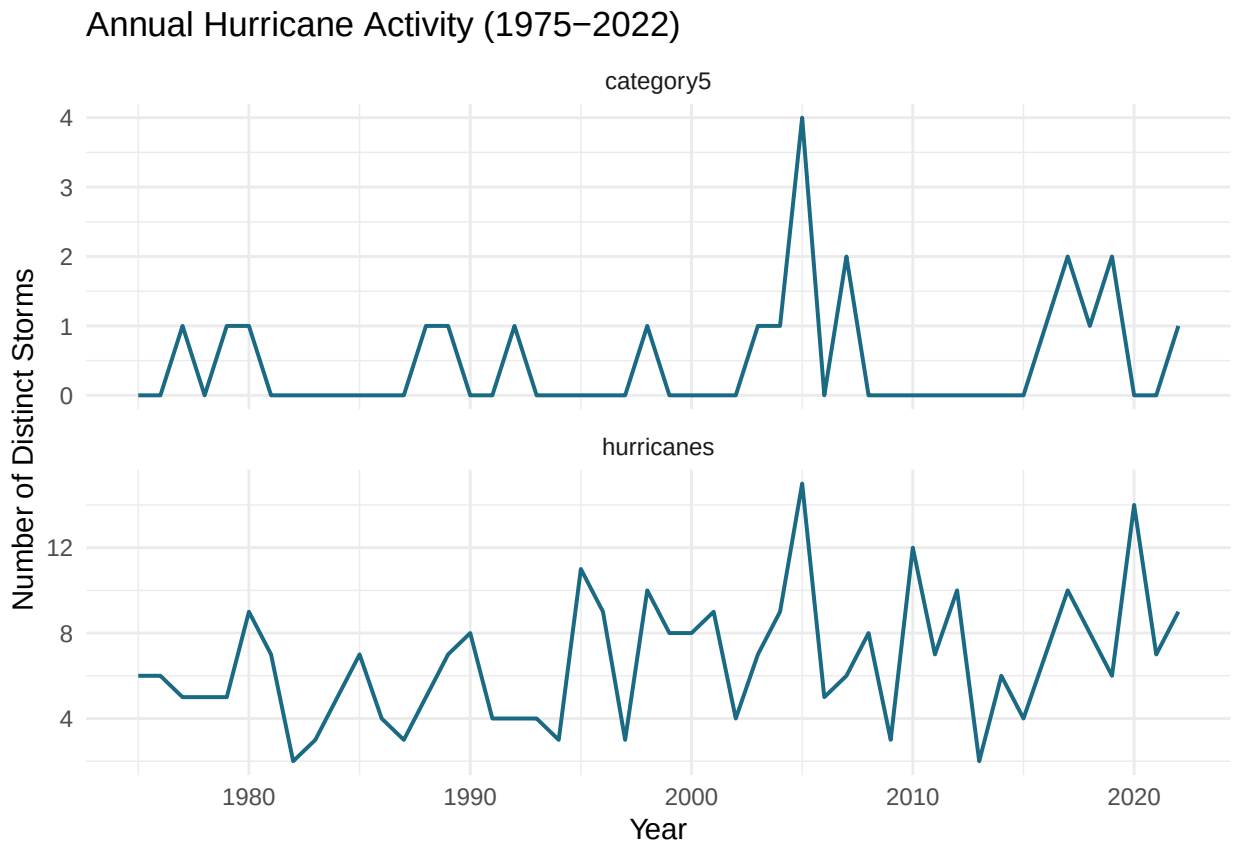
yearly <- annual_hurricane_counts(storms)
print(head(yearly))
```

```
## # A tibble: 6 x 3
##    year hurricanes category5
##   <dbl>      <int>     <int>
## 1  1975          6         0
## 2  1976          6         0
## 3  1977          5         1
## 4  1978          5         0
## 5  1979          5         1
## 6  1980          9         1
```

```r
# Part 3: Hurricane Frequency Visualization
p1 <- yearly |>
  pivot_longer(
```

```
    c(hurricanes, category5),
    names_to = "type",
    values_to = "n"
  ) |>
  ggplot(aes(x = year, y = n)) +
  geom_line(color = "#1B6A84", linewidth = 0.7) +
  facet_wrap(~ type, ncol = 1, scales = "free_y") +
  labs(
    title = "Annual Hurricane Activity (1975-2022)",
    x = "Year",
    y = "Number of Distinct Storms"
  ) +
  theme_minimal()

print(p1)
```




```
# Part 4: Storm-Level Data Cleaning
# Select one peak-wind observation per storm
peak <- storms |>
  filter(!is.na(wind), !is.na(pressure)) |>
  group_by(year, name) |>
  slice_max(
    order_by = wind,
    n = 1,
    with_ties = FALSE
```

```
  ) |>
  ungroup()

stopifnot(nrow(peak) == 655L)
```

```
# Part 5: Custom Function 2
# Convert continuous variables into three groups
to_tercile <- function(x) {
  cuts <- quantile(
    x,
    probs = c(1/3, 2/3),
    na.rm = TRUE
  )

  if (cuts[1] >= cuts[2]) {
    stop("Tercile breakpoints must differ")
  }

  cut(
    x,
    breaks = c(-Inf, cuts, Inf),
    labels = c("Low", "Medium", "High"),
    include.lowest = TRUE
  )
}
```

```
# Part 6: Chi-Squared Test
peak <- peak |>
  mutate(
    wind_group = to_tercile(wind),
    pressure_group = to_tercile(pressure)
  )

frequency <- with(
  peak,
  table(wind_group, pressure_group)
)

chi <- chisq.test(frequency)

print(frequency)
```

```
##            pressure_group
## wind_group Low Medium High
##     Low      0     62  182
##     Medium  33    149   20
##     High   194     15    0
```

```
print(chi)
```

```
##
##  Pearson's Chi-squared test
```

```
##
## data:  frequency
## X-squared = 694.09, df = 4, p-value < 2.2e-16
```

```r
print(min(chi$expected))
```

```
## [1] 62.29618
```

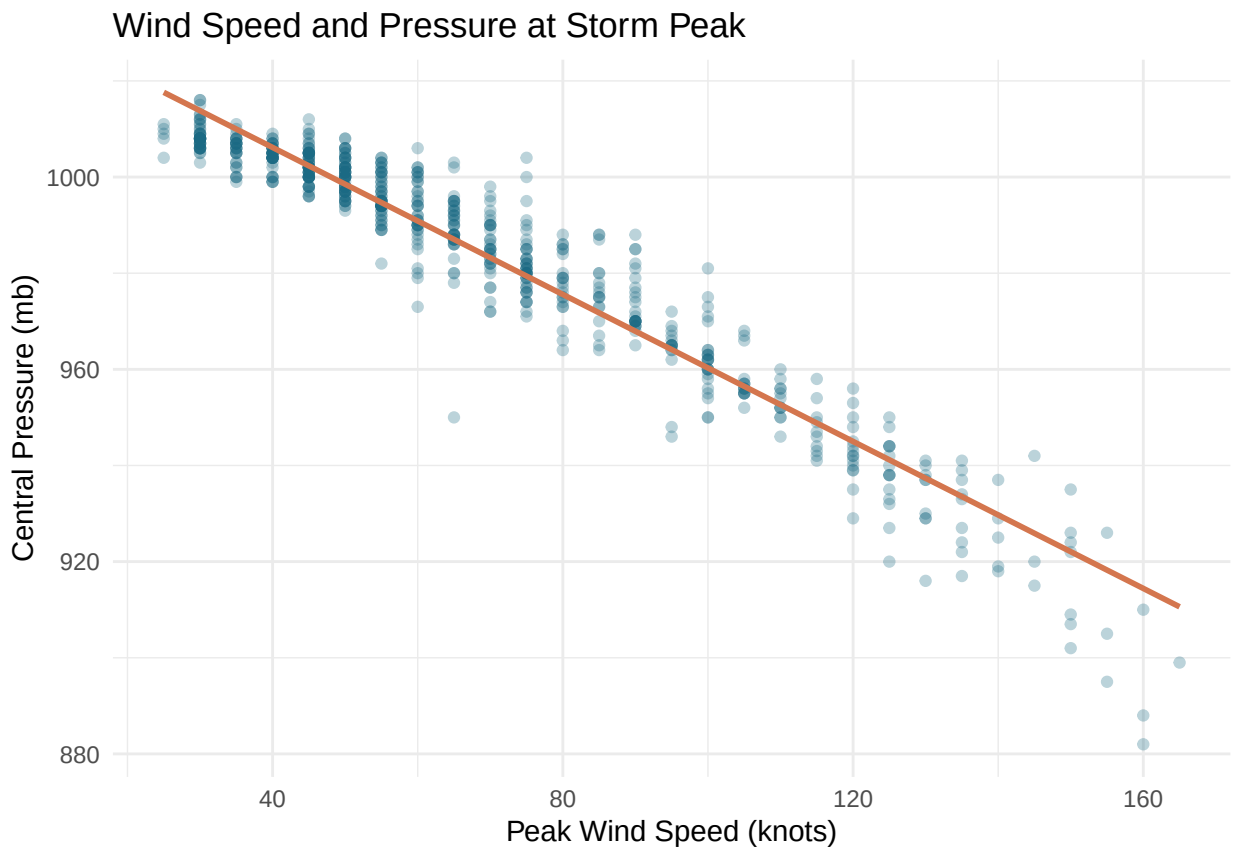
```r
# Pearson correlation
print(cor(peak$wind, peak$pressure))
```

```
## [1] -0.9633473
```

```r
# Part 7: Wind-Pressure Visualization
p2 <- ggplot(
  peak,
  aes(x = wind, y = pressure)
) +
  geom_point(
    color = "#1B6A84",
    alpha = 0.30
  ) +
  geom_smooth(
    method = "lm",
    se = FALSE,
    color = "#D4764E"
  ) +
  labs(
    title = "Wind Speed and Pressure at Storm Peak",
    x = "Peak Wind Speed (knots)",
    y = "Central Pressure (mb)"
  ) +
  theme_minimal()

print(p2)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
# Part 8: Export Visualizations
ggsave(
  "annual_hurricane_counts.png",
  plot = p1,
  width = 7,
  height = 4.2,
  dpi = 300
)

ggsave(
  "storm_peak_wind_pressure.png",
  plot = p2,
  width = 7,
  height = 4.2,
  dpi = 300
)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```